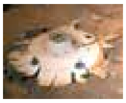
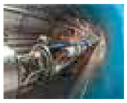


Wreck captured on Mars
NASA's Ingenuity helicopter captures spacecraft wrek on Mars.
https://dgit.in/may22-33



Hadron collider breaks record
The Hadron Collider has started up proton beams at unmatched energy levels after a 3-year shutdown. **https://dgit.in/may22-34**

ming that exposes the cells to those aforementioned Yamanaka factors for only 13 days at which point the cells temporarily lose their identity but permanently lose their age-related markers. After that, the partly reprogrammed cells are given time to grow normally and it was observed that those cells regained their identity (specialisation) and behaved similarly to youthful cells. This research was done specifically with dermal fibroblasts a.k.a skin cells from middle-age donors and showed signs of rejuvenation of around 30 years as measured by a new transcriptome clock.

POTENTIAL APPLICATIONS

For this new method to have real-world applications, they not only need to appear like young cells but also function like them. If that is clinically proven, it can allow faster and more seamless healing of wounds in older people. It can also be used in the treatment of skin conditions like cuts, burns or ulcers. When researchers tested the rejuvenated cells on a layer of cells that had been cut, they found that the new cells moved into the gap faster than the old cells which is a promising result. This same research can open up therapeutic possibilities for the treatment of debilitating diseases like Alzheimer's and cataracts. The method can be applied to other cell types as well such as liver, heart or brain cells which would increase the scope of this kind of research bringing in more potential applications and avenues of treatments.

ALTERNATIVE METHODS

NAD+

Nicotinamide adenine dinucleotide or NAD+ is an essential part of all living cells and is an important part of a plethora of biological processes. The reduction or depletion of NAD+ results in accelerated ageing which can bring about a variety of age-related diseases, metabolic disorders, cancers and neurodegenerative diseases. While the reduction of NAD+ can accelerate ageing, researchers

> **Regenerative biology aims to repair the damage caused by disease or injury to restore function to tissues and organs. As we get older, cells undergo a variety of changes that leads to the gradual loss of their function. As a result, tissues and organs became less functional and the risk of many diseases increases. Regenerative biology can help to reverse these changes and as a result may treat and or prevent age-related diseases.**
>
> ***– Dr Diljeet Gill***
> *(co-author, postdoc, Reik's Lab)*

have found that the elevation of NAD+ levels in cells can not only slow down but also reverse the ageing process which, in turn, can help delay the onset of age-related diseases. Due to that NAD+ has emerged as a promising therapy to battle ageing. NAD+ levels in our cells is not consistent.

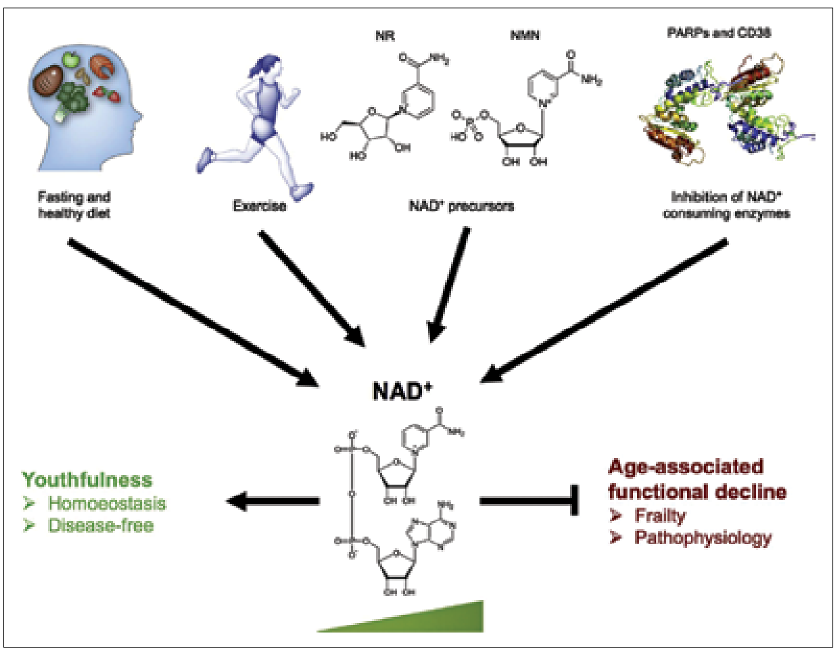


Image Credits: sciencedirect.com

The NAD+ levels in our body reduce with an increase in age. So the idea is to replenish those levels as we grow older to either slow down or in some cases even reverse the ageing process. To increase NAD+ in the body, researchers administer NMN which is converted to NAD+ in the body. NMN, when systematically administered to rodents, enhanced the biosynthesis of NAD+ and the effects could be seen in various tissues like the liver, pancreas, heart, skeletal muscles, kidneys, eyes and blood vessels. Long term administration of NMN has also shown to protect against age-associated functional decline which is evident thanks to an increase in insulin sensitivity, lipid metabolism, energy metabolism, bone density and functions such as the immune system. What's more, long term oral consumption of NMN has shown no obvious side effects or toxic effects. As a whole, an increase in NAD+ with the help of NMN therapy shows a whole lot of promise in reversing the effects of ageing.

But, all the methods we have shown here are still some ways away from being the miracle cure that the current ageing population is looking for. There is far more research to be done to figure out the super long term effects of such treatments so, in the meanwhile, all we can do is sit tight and slow down the effects of ageing in the most natural ways which obviously include a healthy diet and exercise. d